

Annex 3.

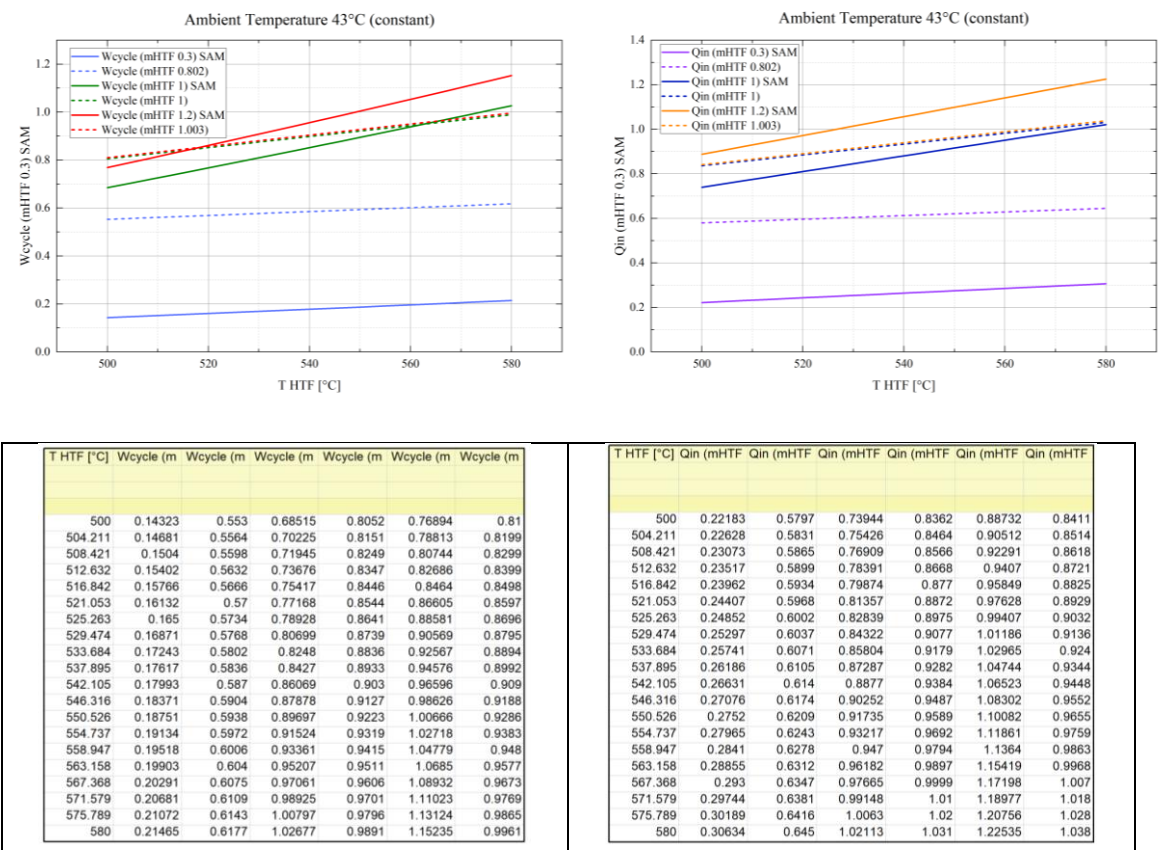
Resultados Paramétricos para la Caracterización del Bloque de Potencia

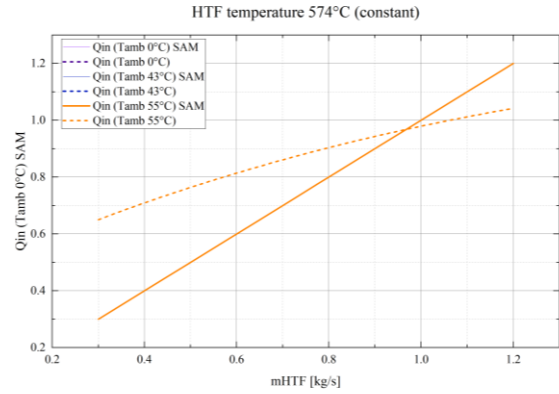
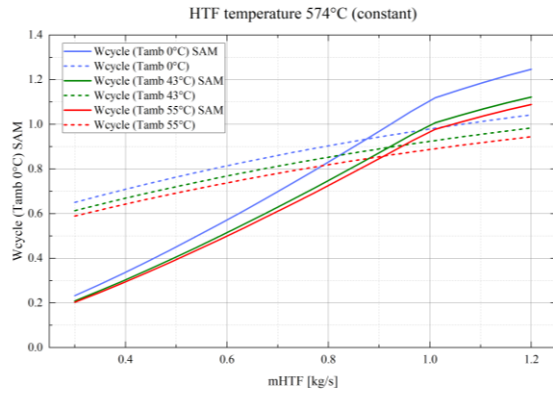
Este anexo presenta un conjunto de resultados paramétricos derivados del modelado termoenergético del bloque de potencia en EES, con base en la formulación propuesta por Wagner (2008) [12]y Wagner & Gilman (2011) [66]. El objetivo principal es caracterizar el comportamiento del ciclo Rankine mediante simulaciones en el entorno System Advisor Model (SAM), considerando variaciones sistemáticas de temperatura y caudal del fluido térmico (HTF), así como de la temperatura ambiente.

Las gráficas muestran el comportamiento de la potencia neta generada por el ciclo (W_{cycle}) y del calor requerido por el bloque de potencia (Q_{in}), ambos valores normalizados respecto a condiciones de referencia. Los escenarios de simulación incluyen:

- Temperatura ambiente constante (43 °C) con variación en T_{HTF} .
- Temperatura del HTF constante (574 °C) con variación en $\dot{m}_{HTF}$.
- Caudal másico constante ($\dot{m}_{HTF} = 1 \text{ kg/s}$) con variación en la temperatura ambiente.

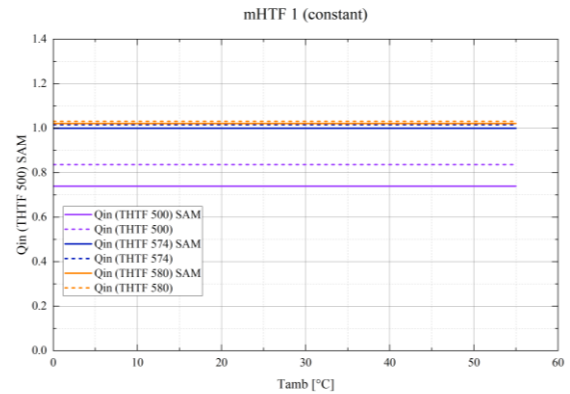
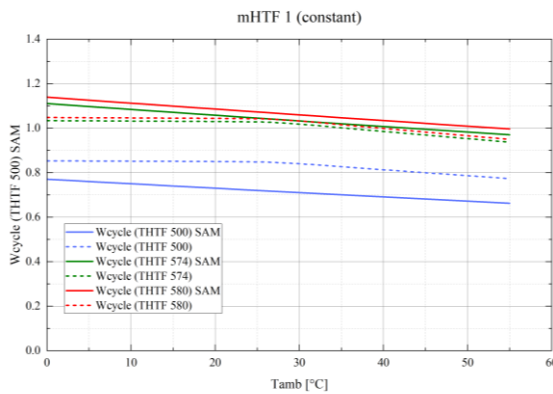
Junto con las gráficas se incluyen las tablas numéricas correspondientes, generadas a partir del modelo del presente estudio.





mHTF [kg/s]	Wcycle (Ta	Wcycle (Ta	Wcycle (Ta	Wcycle (Ta	Wcycle (Ta	Wcycle (Ta
0.3	0.23219	0.6501	0.20905	0.6135	0.20288	0.5892
0.34737	0.28094	0.6787	0.25295	0.6405	0.24548	0.6151
0.39474	0.33173	0.7062	0.29888	0.6684	0.28986	0.64
0.44211	0.38439	0.7326	0.34609	0.6914	0.33587	0.664
0.48947	0.43877	0.7582	0.39505	0.7155	0.38339	0.6871
0.53684	0.49475	0.7828	0.44545	0.7387	0.4323	0.7094
0.58421	0.55224	0.8065	0.49721	0.7611	0.48253	0.7309
0.63158	0.61114	0.8294	0.55025	0.7827	0.534	0.7517
0.67895	0.67139	0.8514	0.60449	0.8035	0.58664	0.7716
0.72632	0.73291	0.8726	0.65988	0.8234	0.6404	0.7908
0.77368	0.79564	0.8929	0.71636	0.8426	0.69522	0.8092
0.82105	0.85954	0.9124	0.7739	0.861	0.75105	0.8269
0.86842	0.92456	0.9312	0.83243	0.8787	0.80786	0.8439
0.91579	0.99065	0.9491	0.89194	0.8957	0.86561	0.8602
0.96316	1.05777	0.9663	0.95237	0.9119	0.92426	0.8758
1.01053	1.1188	0.9828	1.00732	0.9275	0.97759	0.8907
1.05789	1.15414	0.9986	1.03914	0.9424	1.00846	0.905
1.10526	1.1873	1.0137	1.06899	0.9567	1.03744	0.9187
1.15263	1.21814	1.0283	1.09676	0.9703	1.06439	0.9319
1.2	1.2465	1.0422	1.1223	0.9835	1.08917	0.9445

mHTF [kg/s]	Qin (Tamb	Qin (Tamb	Qin (Tamb	Qin (Tamb	Qin (Tamb	Qin (Tamb
0.3	0.3	0.6501	0.3	0.6501	0.3	0.6501
0.34737	0.34737	0.6787	0.34737	0.6787	0.34737	0.6787
0.39474	0.39474	0.7062	0.39474	0.7062	0.39474	0.7062
0.44211	0.44211	0.7326	0.44211	0.7326	0.44211	0.7326
0.48947	0.48947	0.7582	0.48947	0.7582	0.48947	0.7582
0.53684	0.53684	0.7828	0.53684	0.7828	0.53684	0.7828
0.58421	0.58421	0.8065	0.58421	0.8065	0.58421	0.8065
0.63158	0.63158	0.8294	0.63158	0.8294	0.63158	0.8294
0.67895	0.67895	0.8514	0.67895	0.8514	0.67895	0.8514
0.72632	0.72632	0.8726	0.72632	0.8726	0.72632	0.8726
0.77368	0.77368	0.8929	0.77368	0.8929	0.77368	0.8929
0.82105	0.82105	0.9124	0.82105	0.9124	0.82105	0.9124
0.86842	0.86842	0.9312	0.86842	0.9312	0.86842	0.9312
0.91579	0.91579	0.9491	0.91579	0.9491	0.91579	0.9491
0.96316	0.96316	0.9663	0.96316	0.9663	0.96316	0.9663
1.01053	1.01053	0.9828	1.01053	0.9828	1.01053	0.9828
1.05789	1.05789	0.9986	1.05789	0.9986	1.05789	0.9986
1.10526	1.10526	1.0137	1.10526	1.0137	1.10526	1.0137
1.15263	1.15263	1.0283	1.15263	1.0283	1.15263	1.0283
1.2	1.2	1.0422	1.2	1.0422	1.2	1.0422



mHTF [kg/s]	Qin (Tamb	Qin (Tamb	Qin (Tamb	Qin (Tamb	Qin (Tamb	Qin (Tamb
0.3	0.3	0.6501	0.3	0.6501	0.3	0.6501
0.34737	0.34737	0.6787	0.34737	0.6787	0.34737	0.6787
0.39474	0.39474	0.7062	0.39474	0.7062	0.39474	0.7062
0.44211	0.44211	0.7326	0.44211	0.7326	0.44211	0.7326
0.48947	0.48947	0.7582	0.48947	0.7582	0.48947	0.7582
0.53684	0.53684	0.7828	0.53684	0.7828	0.53684	0.7828
0.58421	0.58421	0.8065	0.58421	0.8065	0.58421	0.8065
0.63158	0.63158	0.8294	0.63158	0.8294	0.63158	0.8294
0.67895	0.67895	0.8514	0.67895	0.8514	0.67895	0.8514
0.72632	0.72632	0.8726	0.72632	0.8726	0.72632	0.8726
0.77368	0.77368	0.8929	0.77368	0.8929	0.77368	0.8929
0.82105	0.82105	0.9124	0.82105	0.9124	0.82105	0.9124
0.86842	0.86842	0.9312	0.86842	0.9312	0.86842	0.9312
0.91579	0.91579	0.9491	0.91579	0.9491	0.91579	0.9491
0.96316	0.96316	0.9663	0.96316	0.9663	0.96316	0.9663
1.01053	1.01053	0.9828	1.01053	0.9828	1.01053	0.9828
1.05789	1.05789	0.9986	1.05789	0.9986	1.05789	0.9986
1.10526	1.10526	1.0137	1.10526	1.0137	1.10526	1.0137
1.15263	1.15263	1.0283	1.15263	1.0283	1.15263	1.0283
1.2	1.2	1.0422	1.2	1.0422	1.2	1.0422

mHTF [kg/s]	Qin (Tamb	Qin (Tamb	Qin (Tamb	Qin (Tamb	Qin (Tamb	Qin (Tamb
0.3	0.3	0.6501	0.3	0.6501	0.3	0.6501
0.34737	0.34737	0.6787	0.34737	0.6787	0.34737	0.6787
0.39474	0.39474	0.7062	0.39474	0.7062	0.39474	0.7062
0.44211	0.44211	0.7326	0.44211	0.7326	0.44211	0.7326
0.48947	0.48947	0.7582	0.48947	0.7582	0.48947	0.7582
0.53684	0.53684	0.7828	0.53684	0.7828	0.53684	0.7828
0.58421	0.58421	0.8065	0.58421	0.8065	0.58421	0.8065
0.63158	0.63158	0.8294	0.63158	0.8294	0.63158	0.8294
0.67895	0.67895	0.8514	0.67895	0.8514	0.67895	0.8514
0.72632	0.72632	0.8726	0.72632	0.8726	0.72632	0.8726
0.77368	0.77368	0.8929	0.77368	0.8929	0.77368	0.8929
0.82105	0.82105	0.9124	0.82105	0.9124	0.82105	0.9124
0.86842	0.86842	0.9312	0.86842	0.9312	0.86842	0.9312
0.91579	0.91579	0.9491	0.91579	0.9491	0.91579	0.9491
0.96316	0.96316	0.9663	0.96316	0.9663	0.96316	0.9663
1.01053	1.01053	0.9828	1.01053	0.9828	1.01053	0.9828
1.05789	1.05789	0.9986	1.05789	0.9986	1.05789	0.9986
1.10526	1.10526	1.0137	1.10526	1.0137	1.10526	1.0137
1.15263	1.15263	1.0283	1.15263	1.0283	1.15263	1.0283
1.2	1.2	1.0422	1.2	1.0422	1.2	1.0422